MEENA KANNAN

Lead AI-Augmented Software Engineer | Spring AI, RAG Vector Pipelines, Anthropic MCP Servers & Java/Python Microservices

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PROFESSIONAL SUMMARY

Lead Engineer bridging enterprise Java 21 / Python microservices with production LLM & Agentic architectures. Hands-on creator of Spring AI services, domain-specific RAG vector retrieval pipelines, Model Context Protocol (MCP) backend tool servers, and deterministic evaluation guardrails.

SELECTED IMPACT & ARCHITECTURE HIGHLIGHTS

- Prototyped and rolled out Spring AI, Claude API, and MCP server patterns for backend tool invocation and microservice telemetry.
- Designed RAG retrieval pipelines with custom chunking, vector embeddings, and domain-specific evaluation loops reducing hallucination risk.
- Integrated real-time ML inference REST endpoints with Apache Kafka streams at BlackStraw.AI (15+ mins latency cut to real-time).
- Combined AI anomaly detection models with deterministic Drools BRMS rules at Wolters Kluwer for sub-500ms legal compliance.

CORE EXPERIENCE

Lead Software Engineer | NCS (Client: Land Transport Authority - VRLS 2.0) (Dec 2023 - Present)
Java 21, Spring Boot 3, AWS EKS/ECS, Istio mTLS, Kafka, KEDA, Redis, Angular 18 (-40% redundant API calls), ArgoCD, Spring AI & MCP.

Application Consultant | NCS (Client: LTA VRLS Platform) (Dec 2021 - Dec 2023)
Technical owner of live VRLS platform across 8 licensing modules (Java, Spring MVC, Oracle DB, Kafka, JMS).

Senior Software Engineer | BlackStraw.AI (India) (May 2021 - Nov 2021)
Built Spring Boot + Apache Kafka real-time AI inference pipelines cutting reporting delays from 15+ mins to near real-time.

Software Engineer | HDB Portal Systems (Aug 2019 - Dec 2020) & Wolters Kluwer LegalTech AI (Jul 2018 - Jul 2019)
Engineered sub-500ms AI + Drools BRMS legal billing compliance engine and HDB regulatory transaction workflows.